

Light – Reflection and Refraction

Science • Chapter 9 • Chapter Summary & Study Guide for Daksh

Why does a stick look bent when half-dipped in water? Why does a shaving mirror enlarge your face but a rear-view mirror shows a wide field? Why does a glass slab make a coin underneath look closer? These everyday observations are about **reflection** and **refraction** of light. This chapter develops the rules of these two phenomena — for spherical mirrors and lenses — using sign conventions, formulae and ray diagrams.

Reflection from spherical mirrors

Two laws of reflection (from plane mirrors): angle of incidence = angle of reflection; incident ray, reflected ray, and normal lie in the same plane. These laws apply equally to spherical mirrors. Two types: **concave** (reflecting inner side) and **convex** (reflecting outer side). Important terms:

- **Pole (P)** — geometrical centre of the mirror.
- **Centre of Curvature (C)** — centre of the sphere of which the mirror is a part.
- **Radius of Curvature (R)** = PC; **Focal length (f)** = R/2.
- **Principal Focus (F)** — point on principal axis where parallel rays converge (concave) or appear to diverge from (convex).

Image formation by CONCAVE mirror (real, except when object is between F and P):

Object at infinity	-> at F, real, inverted, highly diminished
Object beyond C	-> between F and C, real, inverted, diminished
Object at C	-> at C, real, inverted, same size
Object between C & F	-> beyond C, real, inverted, magnified
Object at F	-> at infinity, real, very large
Object between F & P	-> behind mirror, VIRTUAL, erect, magnified (e.g. shaving mirror)

Image formation by CONVEX mirror: ALWAYS virtual, erect, diminished
(used as rear-view mirror, gives wider field of view)

Sign convention and the mirror formula

We use the **New Cartesian sign convention**: pole is the origin; principal axis is the x-axis; distances measured along incident light direction are positive (in front of mirror = negative x). Heights above axis are positive, below are negative. Then for any spherical mirror:

Mirror formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$
Magnification : $m = \frac{h'}{h} = -\frac{v}{u}$
(here v = image distance, u = object distance, f = focal length;
 h' = image height, h = object height)

Refraction of light

When light passes from one transparent medium to another, it bends — this is **refraction**. It happens because the speed of light is different in different media (slower in denser media). The two laws:

- Incident ray, refracted ray and normal at the point of incidence lie in the same plane.
- **Snell's law**: $\sin i / \sin r = n_{21}$ (a constant for two given media), where n_{21} is the refractive index of medium 2 with respect to medium 1.

Absolute refractive index $n = c/v$ (speed of light in vacuum / speed in medium). Light bends towards the normal when entering a denser medium and away when entering a rarer one. A glass slab causes **lateral displacement** but no net bending of the emergent ray.

Spherical lenses

A **convex lens** converges parallel light to a point (real focus, +ve f). A **concave lens** diverges parallel light away from a virtual focus (-ve f). Image formation:

- **Convex lens:** real images formed between F and 2F, at 2F, or beyond 2F when object is correspondingly placed beyond 2F, at 2F, or between F and 2F. When the object is *within* F, the image is virtual, erect, magnified — that's how a magnifying glass works.
- **Concave lens:** image always virtual, erect, diminished — formed on the same side as the object, between the optical centre and F.

Lens formula: $1/v - 1/u = 1/f$ (note the MINUS sign)
 Magnification: $m = h'/h = v/u$ (no minus sign for lens)
 Power of a lens: $P = 1/f$ (in metres). Unit: dioptre (D).
 Convex lens: $P > 0$. Concave lens: $P < 0$.

★ MUST-REMEMBER POINTS

- Sign convention: distances measured against the direction of incident light are negative.
- Mirror formula: $1/v + 1/u = 1/f$. Lens formula: $1/v - 1/u = 1/f$.
- Magnification: $m = -v/u$ (mirror); $m = v/u$ (lens). h'/h .
- $f = R/2$ for a spherical mirror.
- Convex mirror always gives virtual, erect, diminished image; concave can give real or virtual.
- Power of a lens $P = 1/f$ (in metres); unit = dioptre (D); convex +ve, concave -ve.
- Light bends towards normal entering a denser medium.

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